

VULNERABILITY ASSESSMENT REPORT

Target Website: <https://lets-chat-delta.vercel.app>

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EXECUTIVE SUMMARY

This report documents a vulnerability assessment conducted on the live web application <https://lets-chat-delta.vercel.app>. The assessment was performed using industry-standard security tools including Nmap, OWASP ZAP, cURL, Browser Developer Tools, and manual inspection techniques. The objective of this assessment was to identify security misconfigurations, exposed services, missing security headers, and other web security issues that could potentially affect confidentiality, integrity, and availability.

The assessment was conducted using passive and non-intrusive testing methods only. No exploitation or destructive actions were performed during the assessment.

ASSESSMENT SCOPE

- Target URL: <https://lets-chat-delta.vercel.app>
- Assessment Type: Passive Vulnerability Assessment
- Assessment Date: May 2026
- Tools Used: Nmap, OWASP ZAP, cURL, Chrome DevTools
- Testing Approach: Reconnaissance, Passive Scanning, Manual Inspection

ASSESSMENT METHODOLOGY

Reconnaissance with Nmap

Nmap was used to identify active services, open ports, and network exposure. The scan confirmed that the target application exposes HTTP and HTTPS services on ports 80 and 443.

HTTP Header Inspection using cURL

cURL verbose mode was used to inspect HTTP response headers, SSL/TLS configuration, server behavior, caching policies, and transport security settings.

Browser Developer Tools Analysis

Chrome DevTools Network tab was used to analyze API requests, login requests, status codes, and frontend resource loading.

OWASP ZAP Passive Scan

OWASP ZAP passive scanning was performed to identify missing security headers, cookie-related issues, caching problems, and other web application security weaknesses.

DETAILED FINDINGS

Missing Content Security Policy (CSP) Header

Risk Level: Medium

OWASP ZAP detected that the application does not define a Content-Security-Policy header. Without CSP protection, the application becomes more vulnerable to Cross-Site Scripting (XSS) attacks.

Remediation: Implement a strict CSP policy to restrict execution of untrusted scripts and resources.

Missing X-Content-Type-Options Header

Risk Level: Medium

The application does not include the X-Content-Type-Options header. This may allow MIME type sniffing attacks by browsers.

Remediation: Add X-Content-Type-Options: nosniff to all HTTP responses.

Missing Anti-Clickjacking Header

Risk Level: Medium

The website does not include X-Frame-Options or equivalent frame protection headers. Attackers may attempt clickjacking attacks using malicious iframes.

Remediation: Configure X-Frame-Options DENY or SAMEORIGIN.

Information Disclosure through Headers

Risk Level: Low

Response headers reveal server and caching information including Vercel infrastructure details.

Remediation: Limit unnecessary information disclosure in HTTP headers where possible.

NMAP SCAN RESULTS

The Nmap scan identified the following exposed ports and services:

Port	State	Service
80/tcp	Open	HTTP
443/tcp	Open	HTTPS

ASSESSMENT SCREENSHOTS



Figure 1: Nmap Scan Output



Figure 2: cURL SSL/TLS Verification



Figure 3: Browser Developer Tools Network Analysis

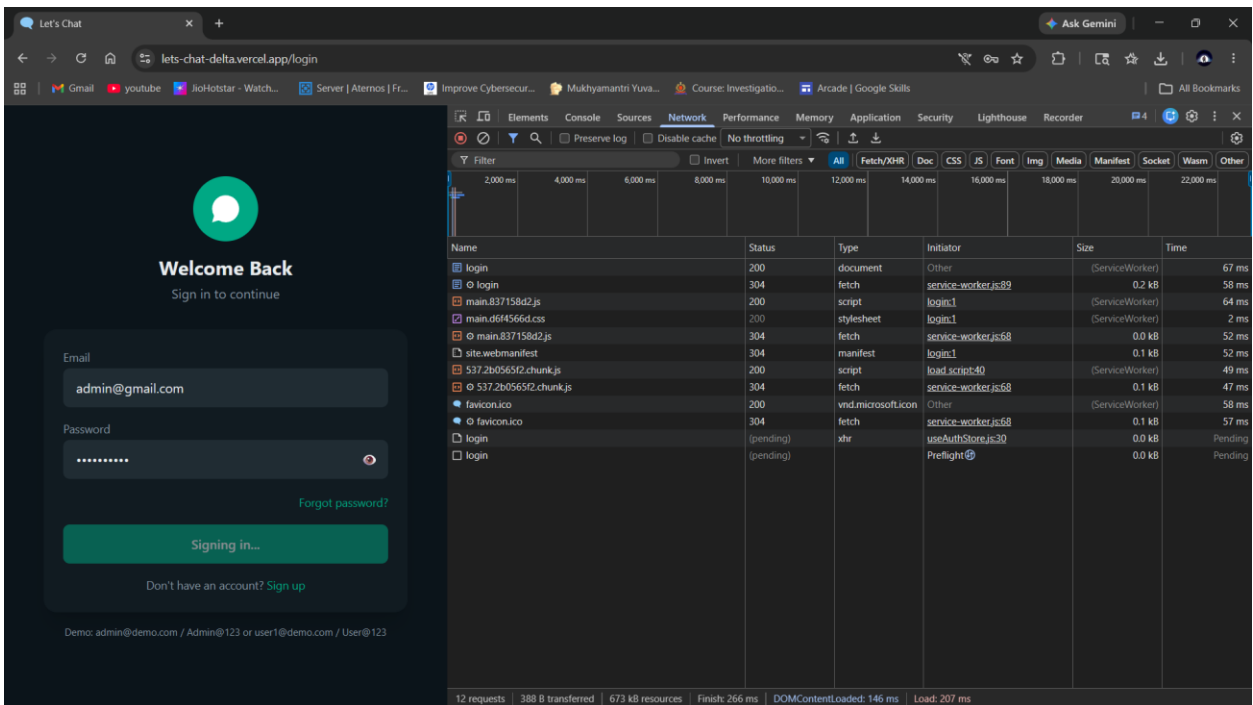


Figure 4: OWASP ZAP Passive Scan Findings

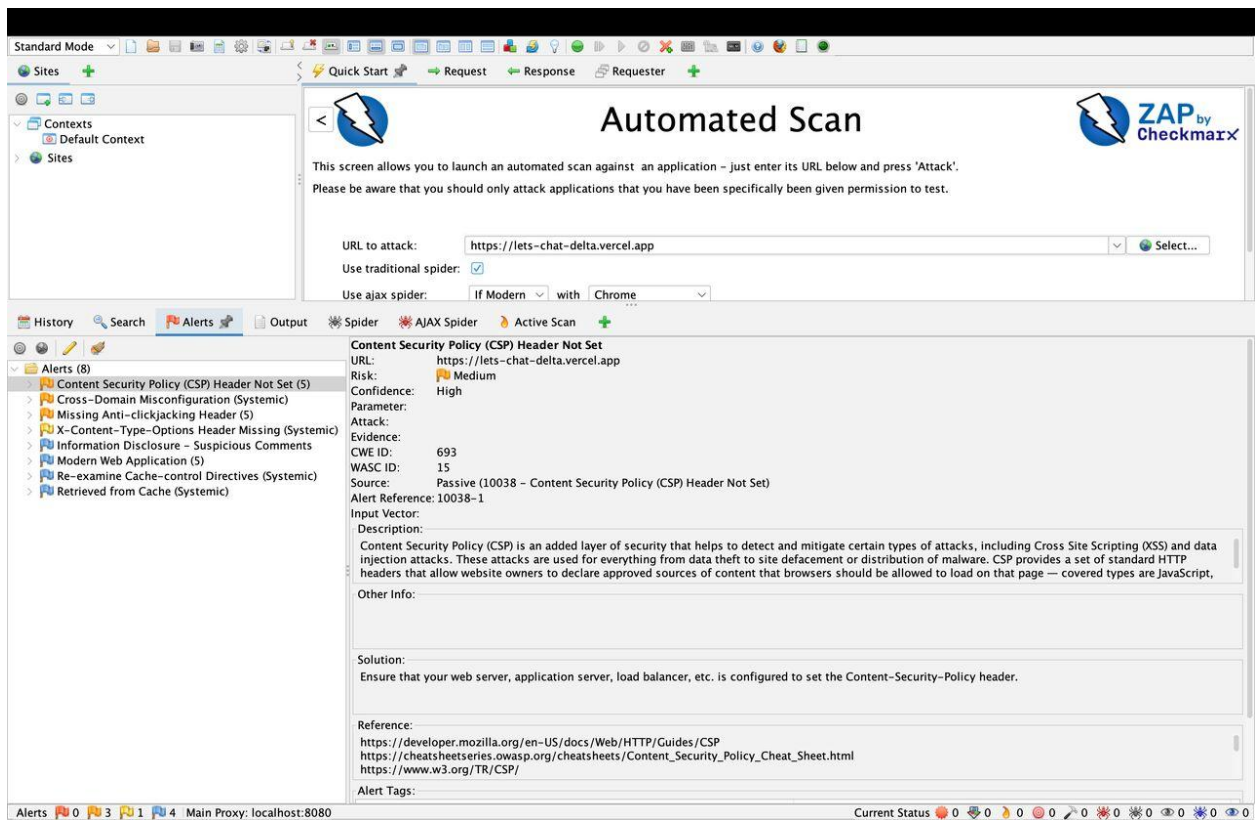


Figure 5: Login Interface of Target Application

RECOMMENDATIONS

- Implement a strong Content Security Policy (CSP).
- Add all recommended HTTP security headers.
- Enable secure cookie flags including HttpOnly and Secure.
- Regularly perform vulnerability assessments and penetration testing.
- Monitor logs and suspicious activities continuously.
- Apply security hardening and dependency updates regularly.

CONCLUSION

The assessment identified several medium and low-risk vulnerabilities related to security headers and application configuration. While no critical vulnerabilities were exploited during this assessment, implementing the recommended mitigations will significantly improve the overall security posture of the application. Regular assessments, patch management, and secure development practices are strongly recommended.